

CERRO GORDO COUNTY
AMATEUR RADIO EMERGENCY SERVICE

STANDARD OPERATING PROCEDURES

Revised July 2026

Emergency Coordinator: Justin Bonjour, N0JBE

Cerro Gordo County, Iowa

1. Introduction and Mission

Cerro Gordo County ARES (CGARES) is a volunteer group of licensed amateur radio operators dedicated to providing reliable emergency and public service communications when normal systems fail. We support local emergency management, public safety agencies, and the community during disasters, severe weather, large events, and other incidents.

Our focus is on **preparedness, training, and effective response**. We maintain a small, practical, and professional team that emphasizes interoperability with Iowa ARES, the National Traffic System, and served agencies.

Emergency Coordinator (EC): Justin Bonjour, N0JBE

2. Core Communications Plan

This section lists the frequencies and modes that every CGARES member should have programmed and be comfortable using. These are the primary tools we train on and use during operations.

VHF / UHF Voice

Priority	Purpose	Frequency	Offset / Tone	Mode	Notes
Primary	Main Repeater Net	147.15 MHz	+600 kHz, TSQ 103.5 Hz	FM Voice	Main weekly net frequency (except 1st Wednesday)
Secondary	Alternate Repeater	442.275 MHz	+5.0 MHz, Tone 100.0 Hz	FM Voice	Sunday 6:30 PM net
Tactical	Primary Simplex	147.570 MHz	Simplex	FM Voice	1st Wednesday simplex net & county-wide calling
Tactical	Alternate Simplex	147.405 MHz	Simplex	FM Voice	Backup tactical / simplex channel

Repeater Failure Plan: If the primary repeater (147.15 MHz) fails, move to **147.570 MHz simplex**.

Digital & Data Modes (VHF/UHF)

Purpose	Frequency	Mode	Notes
Primary Digital Data	145.050 MHz	1200 baud Packet / Winlink	Primary frequency for Winlink and packet operations
Tactical Digital	145.050 MHz	MT63	Robust tactical messaging (FLDigi)
Local ARES APRS	144.990 MHz	1200 baud APRS	Local practice, messaging, and position tracking
National APRS	144.390 MHz	1200 baud APRS	Standard North America APRS frequency

HF / NVIS (Regional Communications)

These frequencies provide reliable Near Vertical Incidence Skywave (NVIS) coverage across Iowa and into neighboring states. Use primarily during Echo 2 or higher, or when directed by the EC for regional traffic handling (health & welfare, coordination with state resources, etc.).

Purpose	Frequency	Mode	Notes
Primary HF Voice	3.970 MHz	LSB	Iowa Traffic & Emergency Net (ITEN) – statewide interoperability
Secondary HF Voice	7.250 MHz	LSB	Daytime alternative on 40m
Primary HF Digital	3.5885 MHz	USB – Olivia 8/500 (Iowa ARES Digital Net)	Iowa ARES Digital Net (NBEMS) – starts on Olivia 8/500, moves to THOR 22 when conditi
Secondary HF Digital	7.104 MHz	USB – THOR 22 or Olivia 8/500	Alternate frequency and mode

Digital Mode Guidance: Primary HF Digital Mode is **Olivia 8/500** (most robust, matches Iowa state net). Secondary is **THOR 22** (faster when conditions allow). You may switch modes during a net if conditions justify it. Olivia 8/500 is the default when in doubt.

3. Echo Scale and Activation Levels

The Echo Scale is CGARES's standardized readiness and threat level indicator. It tells members how seriously to pay attention and what posture to maintain.

Important Notes:

- The Echo level is **completely separate** from whether a net is running. A Skywarn net, resource net, traffic net, or any other net can be active at any Echo level.
- Even a severe thunderstorm warning or tornado warning in the county does **not** automatically raise us to Echo 2. We will often remain at Echo 3 during most warnings.
- Members should be able to look at current conditions (weather, radar, local events, etc.) and reasonably determine the current Echo level themselves.

Detailed Echo Scale

Echo 5 – Normal / Routine Operations

No significant threats or watches in the county.

Expected Actions: Normal casual monitoring. Check the primary repeater when convenient. No special preparations required.

Echo 4 – Elevated Alert

Weather watch issued or potential county event that could require communications support. Increased traffic on the repeater is expected.

Expected Actions:

- Keep a go-kit accessible and ensure batteries are charged.
- Periodically monitor the primary repeater during waking hours.
- Every 4–6 hours, announce your call sign on the repeater (example: "N0JBE monitoring"). This is a quick 5-second check-in.
- Check Winlink at least once daily.

Echo 3 – Increased Monitoring

A warning has been issued for Cerro Gordo County or a confirmed local hazard exists.

Expected Actions:

- Keep your radio on and monitoring the primary repeater while awake.
- Announce your call sign on the repeater every 2–3 hours while awake (example: "N0JBE monitoring").
- Check Winlink at least every 4 hours while awake.
- Be prepared to respond promptly if called by name on the repeater.

Echo 2 – Deployment Likely

Situation has developed where ARES deployment appears likely based on weather conditions, ground reports, or a request from served agencies. Limited support operations may already be underway.

Expected Actions:

- Maintain continuous monitoring of the primary repeater whenever practical.
- Monitor assigned digital modes as practical.
- Be prepared to deploy with your go-kit on short notice.
- A Resource Net will typically be stood up.

Echo 1 – Deployment

ARES is actively supporting an incident or event. This may occur due to a formal request from Emergency Management or other served agencies, or when the Emergency Coordinator determines ARES support is needed (such as supporting National Traffic System traffic).

Expected Actions: Respond as directed by Net Control or the Emergency Coordinator. Deploy with your go-kit if requested. Follow all safety and operational guidance.

Echo 0 – Recovery / Hot Wash

Post-event debrief and lessons learned review.

Expected Actions: Participate in the after-action review when possible and submit any lessons learned.

4. Station Status Reports (STATREP)

The **Station Status Report (STATREP)** is a quick, standardized way for members to report their current operational readiness using a simple color-coded system. This allows the Emergency Coordinator to rapidly assess team capabilities without long conversations on the air.

Purpose

STATREPs give leadership a clear snapshot of who is available, what resources they have, and any limitations. They are especially useful during elevated readiness (Echo 4 and above) and during monthly training.

Color Code Definitions

- **Green** — Fully operational and ready for deployment.
- **Yellow** — Partially degraded (limited availability, minor equipment issues, or short on resources).
- **Red** — Not operational or unavailable at this time.

Typical Fields to Include

Station / Call Sign • Status Color (Green / Yellow / Red) • Personnel: OK / Short X • Fuel / Power: OK / Low / Critical • Comms (radios, antennas, digital): OK / Issues • Equipment / Go-Kit: OK / Needs... • Remarks (optional short notes)

How to Send a STATREP

- **Voice:** Read it plainly and concisely when Net Control requests it during the net.
- **FLmsg:** Use the custom STATREP template (available from the EC). This produces a clean, standardized form that can be sent over FLDigi.
- **APRStac:** Use APRStac status beacon feature if your station is configured for it. This allows automatic or periodic broadcasting of your status.

STATREPs are typically sent on the **2nd Wednesday** of each month as regular traffic during the repeater net. Members may send their STATREP via voice, FLmsg, or APRStac — whichever method works best for them.

5. Resource Nets

A **Resource Net** is a quick, directed net used to inventory available operators and resources. Net Control asks for short reports on: who is available (and for how long), go-kit status, mobile capability, extra equipment (batteries, generators, digital modes, etc.). Check-ins are kept very short (~30 seconds per person).

Resource Nets can be announced on an existing Skywarn net (“We’re switching to a resource net for the next few minutes…”) without changing frequency, then switched back to regular net operations when complete.

6. Net Procedures

Weekly Nets

- **Every Wednesday at 7:00 PM Central Time**

– 1st Wednesday: Simplex net on 147.570 MHz

– All other Wednesdays: Repeater net on 147.15 MHz (TSQ 103.5 Hz)

- **Sunday 6:30 PM:** Alternate repeater net on 442.275 MHz

How to Check In

When Net Control asks for check-ins, give your **call sign phonetically, name, and location** (city or area).

Example: “N0JBE, November Zero Juliet Bravo Echo, Justin, Mason City.”

Recommended Monthly Training Focus

To keep training consistent and predictable, members are encouraged to prepare and send the following items as regular traffic during the appropriate week. When Net Control asks for stations with traffic, simply offer your prepared item the same way you would any other message.

- **1st Wednesday (Simplex Net):** Weather Report using the four-part format (Temperature, wind direction/speed, sky conditions/precipitation, any hazards).
- **2nd Wednesday:** Station Status Report (STATREP) – Green/Yellow/Red format.

- **3rd Wednesday:** Radiogram (send to anyone about anything; practice the proper radiogram format).
- **4th Wednesday:** Regular net (no specific training item).
- **5th Wednesday (when it occurs):** ICS-213 drill (a specific message will be provided by the EC the week prior).

Training items are sent as regular traffic. There is no separate "training traffic" block.

FLDigi Monitoring: The Net Control Station (NCS) monitors FLDigi on the primary digital NVIS frequency of **3.5885 MHz** during nets.

7. Digital Operations

We use the following tools for resilient communications:

- **FLDigi** — Primary digital mode software (Olivia 8/500, THOR 22, MT63, etc.)
- **Winlink** — For sending forms, lists, and detailed reports when internet is unavailable
- **APRStac** — Position tracking, messaging, and situational awareness (use APRStac for station status when possible)
- **Packet Radio** — On 145.050 MHz for Winlink and basic messaging

Training Expectation: New members are expected to become proficient in FLDigi and basic APRStac within a reasonable time. Winlink is encouraged but not mandatory for all members.

8. Emergency Operations Center (EOC)

Our primary coordination hub is the **Cerro Gordo County EOC**, located at **2716 S. Federal Ave., Mason City, IA**. We stage here during activations when appropriate and adapt to the situation as needed.

9. Common Pro Words

Use these to keep communications clear and professional:

- **Roger** — Message received and understood
- **Wilco** — Will comply
- **Affirmative** — Yes
- **Negative** — No
- **Over** — End of transmission; response expected
- **Out** — End of transmission; no response expected
- **Say Again** — Repeat your last transmission
- **Break** — Pause for priority traffic or to separate message parts
- **Figures** — Numbers follow
- **Correction** — Error in previous transmission; correct info follows
- **Stand By** — Wait briefly
- **Clear** — Ending the net or QSO

10. Training and Readiness

- Participate in weekly nets regularly.
- Maintain your go-kit, radios, antennas, and backup power in ready condition.
- Practice digital modes (FLDigi, APRStac, Winlink) during non-emergency times.
- Know the Echo Scale and be able to self-assess your readiness level.
- Prepare and send the recommended training item for the week as regular traffic during the net (see Net Procedures section).

11. Contact Information

Emergency Coordinator: Justin Bonjour, N0JBE

Questions or activation requests: Contact the EC directly or via the primary repeater (147.15 MHz).

This SOP is designed to be clear, practical, and professional while remaining realistic for volunteer operators in North Iowa. It balances readiness with respect for people's time and equipment limitations.

73,

Justin N0JBE

Emergency Coordinator – Cerro Gordo County ARES